Contribution Concentration and Maintenance Outcomes: A Longitudinal Case Study of Headscale

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Working draft, 15 October 2026

Abstract

Repository activity, maintenance responsibility and repair outcomes describe different phenomena. We investigate these distinctions in Headscale, a self-hosted implementation of the Tailscale control server. Our reproducible case study combines release-pinned Git history, 1371 issues, a separate census of 2166 pull requests (PRs), a complete submitted-review and lifecycle census, a comment-process sample of 76 PRs, and evidence for 100 issues. We examine author identity, file renaming, closure mechanisms and competing PR outcomes before interpreting maintenance trends. Linking validated core-contributor aliases changes production-code author concentration in 2026 Q2 from Herfindahl–Hirschman index (HHI) 0.536 to 0.920. The issue evidence sample contains 29 automated final closures and only 5 explicit closing-commit links, with component checks supporting 2 local repair links. Closure alone cannot establish repair. On the same 87 merged PRs in 2026 Q2, author, reviewer and integrator HHIs are 0.545, 0.388 and 0.700; review credit is shared per PR. Of 783 self-merges, 761 have an eligible other-account submitted review. Reopening matters: 31 PRs initially close but subsequently merge. An exploratory cause-specific model of initial processing episodes estimates a merge hazard ratio of 0.998 per 0.1 increase in prior integration HHI (95% interval 0.927–1.074). Later-period associations are sensitive to calendar-cohort control. These measurements support concentrated activity and heterogeneous closure mechanisms, without establishing a causal scaling crisis. Independent defect adjudication and explanatory mechanism validation remain pending. The case motivates separating activity, responsibility and outcome boundaries when evaluating vendor-adjacent open-source maintenance.

Keywords: Empirical software engineering; open-source maintenance; contribution roles; pull requests; issue closure; mining software repositories.

1 Introduction

Open-source infrastructure can attract many contributors while remaining dependent on a small maintenance core. Yet contributor growth alone does not establish that review, integration and defect handling have become more distributed. Conversely, concentrated contributions can reflect specialised expertise rather than declining capacity. An empirical account needs to distinguish these competing interpretations and establish what its measurements actually observe.

Headscale provides a useful context for this problem. It implements an open-source, self-hosted Tailscale control server. Its stated scope centres on self-hosters, hobbyists and smaller organisations, rather than unrestricted parity with the commercial service [9]. A maintainer's FOSDEM presentation describes the project's growth, upstream funding and protocol relationship with Tailscale [3]. We use that account as contextual evidence; we do not treat it as independent verification of repository-derived conclusions. The upstream client, deployment

configuration and server implementation also create distinct responsibility boundaries for reports filed in one issue tracker.

The motivating pilot observed growth in nominal contributors, a late increase in commit concentration and longer issue-closure durations. Such a pattern raises a maintenance question, but cannot answer it by itself. File modifications need not correspond to coordination handoffs; a closed bug-labeled issue need not represent repaired server code; and multiple author emails need not identify different people. We therefore audit the measurements before developing an explanation of the project's trajectory.

We provide matched comparisons of author, reviewer and integrator activity under different artifact scopes; time-ordered models of initial PR processing episodes; and traceable issue evidence separating closure from supported repair. Release-pinned reproduction and identity/path audits support these measurements. The results describe this case; the proposed explanation connecting maintenance roles with client, server and deployment boundaries still requires independent semantic validation.

2 Related work and analytical framing

Headscale has already been evaluated as a research subject. Kjorveziroski et al. compare it with Netbird and ZeroTier in a full-mesh VPN performance study, measuring throughput and Kubernetes API response under adverse network conditions [22]. Their question concerns deployed-system performance, whereas ours concerns longitudinal repository maintenance. Selecting Headscale therefore does not itself supply novelty.

Machairas's DevSkill thesis includes Headscale in a trending-repository illustration (p. 37), while its questions concern aggregate developer skills and technology trends [25]. It also explicitly identifies email/noreply identity splitting (p. 58). Our case-specific alias correction quantifies an established validity problem; it does not introduce that problem.

Code ownership has long been examined in relation to software quality. Bird et al. studied ownership and failures in industrial systems, showing that ownership is an empirical construct requiring specific operationalisation and validation [1]. Their results do not justify interpreting every concentration statistic as knowledge loss or maintenance debt. Research on coordination breakdowns similarly concerns coordination requirements and observed process failures, rather than the elapsed time between unrelated edits by different authors [2].

Jabrayilzade et al. combine reviews, meetings and version-control activity for bus-factor estimation and validate against engineer assessments on 13 projects [18]. Their discussion already identifies senior reviewers who contribute little authored code. Our observed author/reviewer discrepancies illustrate that established concern; they do not introduce it. Without knowledge validation or meeting evidence, our activity HHIs are not validated bus-factor estimates.

Separating contribution roles is established prior work. Gousios et al.'s survey of 749 integrators identifies quality assurance and prioritisation as central integration challenges [4]. Thongtanunam et al. explicitly distinguish authoring from reviewing activity and examine review-aware ownership across Qt and OpenStack releases in relation to software quality [27]. Our matched role measurements therefore apply an established distinction to this case; they do not introduce review-aware ownership. Lulla et al.'s recent preprint studies declared code owners and review dynamics across more than 844,000 PRs [24]. Our single-project description of declared owners cannot establish effects of CODEOWNERS adoption or compete with that study's population scope.

PR responsiveness is also an established topic. Hasan et al. distinguish bot-generated and human-generated first responses [5]. Khatoonabadi et al. model maintainer and contributor response latency across multiple projects [21]. Our response definitions exclude author self-comments and bots and explicitly terminate the response opportunity at disposition. We distinguish a response from a completed submitted review and avoid inferring historical maintainer

status from a present-day account association field. Administrative automation is also an established subject. Khatoonabadi et al. study Stale bot and PR workflow across 20 projects [20]. Our observation of automated closures is a case measurement audit, not a new discovery of Stale bot or evidence of its causal effects. Issue classification is likewise an established data-validity concern. Herzig et al. manually examined reports from five projects and demonstrated that treating non-defect changes as bugs biases prediction studies [17]. Our pending responsibility/outcome coding therefore needs to validate a case-specific explanatory claim; identifying misclassification alone would not supply a novel contribution.

Recent PR-workflow mortality research examines repository decline at a much larger scale [19]. Headscale is an active case, so we do not classify it as failed merely because some indicators become concentrated or slower. Our intended explanatory contribution concerns role differentiation and responsibility boundaries in an upstream-dependent implementation. Its novelty must be established through validated process evidence, not by the use of HHI or the popularity of the selected repository. The remaining candidate contribution is an explanatory account of how upstream-client and deployment responsibility boundaries interact with maintenance decisions in a deliberately scoped implementation. The present event audit and illustrative cases motivate that account, but do not yet validate it across the independently coded sample.

Broader OSS-health assessment already incorporates organisational context. Linåker et al. combine interviews with 17 practitioners, comparison with existing frameworks and an industrial case application [23]. Their findings explicitly consider governance concentration and contextual differences. Our role measurements and contextual interpretation therefore do not constitute a new general health-assessment framework. A distinct contribution would require showing how traceable responsibility decisions change the interpretation of maintenance outcomes in this implementation.

3 Case study design

We follow the principle of studying software engineering phenomena in their context and retaining a traceable chain of evidence [26]. The case is Headscale maintenance, with embedded units consisting of Git changes, PR dispositions and issue closure episodes. We ask:

1. RQ1: How does concentration differ across contribution, review and merge roles over the project's history?

2. RQ2: What relationship can be established between prior maintenance concentration, workload and subsequent response or disposition outcomes?

3. RQ3: Which observable closure and responsibility mechanisms change the interpretation of repository-level maintenance indicators?

RQ1 has completed contribution, review and integration measurements, with pilot comment evidence. RQ2 has completed exploratory association models, whose assumptions and scope are reported below. RQ3 has completed event extraction, illustrative case evidence and two bounded component checks; independent defect-outcome coding remains pending.

The release-pinned contribution policy requires advance discussion of new features and a plan for their subsequent maintenance. It explains this model in terms of review burden and burnout prevention [8]. The declared owner file names two default owners and additional documentation and infrastructure owners [10]. These documents establish declared governance practices; they do not establish branch-rule enforcement, historical compliance or a causal effect of the policy.

Competing explanations. The post-pilot design considers workload bottlenecks, deliberate role specialisation and responsibility routing as alternative explanations. Different author/reviewer/integrator concentrations alone distinguish observed activity, not the reasons for that activity. A scope-routing explanation requires a time-ordered link between evidence about the affected component, an explicit maintenance decision and the observed treatment or closure. Client and deployment labels alone cannot establish that mechanism. Contradictory cases and unclear decisions must remain in the analysis; the current illustrative cases motivate, rather than complete, this test.

3.1 Data sources and observation boundaries

Git analysis ends at the immutable commit underlying stable release v0.29.5, published on 13 October 2026 [11]. We retain the earlier v0.28.0 endpoint as a baseline. An independent bare repository prevents edits in the development checkout from entering measurements. Source manifests record commit hashes, script hashes, timestamps and known limitations.

The completed issue list includes all states and labels, rather than only closed bug reports. It contains 1371 issues, of which 773 have the bug label at collection time. A separate completed PR list contains 2166 PRs, including 1667 human-authored submissions. The two API snapshots were collected at different times and are not asserted to be an atomic census at the release timestamp. API pagination completeness and page hashes are checked independently of endpoint content validity.

PR account classification treats GitHub Bot accounts and logins ending in `[bot]` as automated. The remaining class identifies accounts, not proof that all their content was manually authored; user accounts can also submit automated or AI-assisted changes.

The PR process pilot samples up to twelve PRs per creation year with a fixed random seed, yielding 76 selected PRs. Every selected PR has complete detail, review, issue-comment and inline-comment endpoints. Inclusion probabilities are retained. This design requires sampling weights before population estimates and provides limited precision within individual quarters. The issue validation sample contains 100 cases stratified by creation year, observed state and observed closure reason. The sample oversamples some closure categories; its raw mechanism counts are not population rates.

3.2 Operational definitions

For activity counts c_i, define $s_i = c_i / \sum_j c_j$ and

$$H = \sum_i s_i^2, \qquad N_{\text{effective}} = 1/H.$$

These measure concentration and equivalent equally active identities. They do not measure knowledge ownership. We name the minimum number of contributors accounting for half of activity literally, rather than calling it a validated bus factor. Fixed calendar quarters replace unequal lifecycle segments for trend descriptions; boundary quarters are excluded from volume comparisons. Production Go, tests, generated files and documentation are separated.

Raw email identities are contrasted with available account links validated through GitHub attribution of representative commits. This validation covers core aliases, not every contributor. Git's inferred rename edges form a path sensitivity analysis. They do not establish identity across file splits, reuse or low-similarity moves. Historical path groups are checked against files present at the endpoint before they are described as current artifacts.

Issue duration runs from creation to the latest observed closure or the observation cutoff. Open issues are right-censored. We report time to closure, not MTTR. Current labels and state cannot reconstruct historical eligibility, and reopened episodes require additional timeline

handling. A closure linked to a commit is stronger evidence than a closure timestamp, but still requires examination of whether the change addresses the reported behaviour.

PR disposition has three observed outcomes: merge, closure without merge, and open censoring. Closure without merge is a competing event, not necessarily rejection. First human response is the earliest eligible review or comment by someone other than the author. A PR merged or closed before such a response has a separate competing outcome. Later comments do not retrospectively extend the earlier waiting opportunity. All pilot PR timelines have been collected, allowing the first disposition to terminate the initial response episode. The pilot includes 7 PRs with reopening events and 12 with draft-state history. The primary response clock starts at creation, rather than readiness for review, so it is not a measure of exclusively active review waiting.

4 Completed descriptive results

4.1 Activity concentration is sensitive to author identity

The cumulative legacy comparison grows from 2803 commits and 238 author names at v0.28.0 to 3401 commits and 259 author names at v0.29.5. The top-author share rises from 48.16% to 54.78%. These quantities exclude the initial release history in subsequent cumulative ranges and use the legacy author-time and name rules; the research series uses full endpoint ancestry and committer time.

Email identities introduce a separate distortion. The core maintainer appears under three email identities, all attributed by GitHub to the same account. Four of five core-alias candidates receive account attribution; one founder alias remains unlinked. In production Go changes during 2026 Q2, author HHI is 0.536 under email counting and 0.920 after available core links. Replacing committer time with author time yields 0.920 for the linked series in the same quarter, so this particular concentration result persists under the timestamp sensitivity. Figure 1 shows the identity sensitivity over complete quarters. The result supports concentrated authorship under the validated rule. It does not quantify how much knowledge, reviewing authority or operational responsibility the maintainer possesses.

4.2 Historical paths and current artifacts are different units

Git history yields 844 inferred rename events. The historical `app.go` activity group maps to current `hscontrol/app.go`; its activity does not stop merely because the old path stops appearing. Other highly active historical groups have no live representative at the endpoint. Therefore a high historical rank is not enough to establish a present structural bottleneck. Figure 2 reports raw-path and rename-linked concentration within quarters. Neither series includes structural complexity or coupling; it is inappropriate to call either a direct measure of technical debt.

4.3 Contribution and integration are distinct observed roles

All 1318 merged human-authored PRs have recorded merger accounts. Across the observation period, 5 accounts perform these actions, and 783 PRs are merged by their own authors. This is an integration-action census, not an estimate of all review work or all knowledge holders. Direct pushes and upstream coordination are outside its population.

To compare roles without changing PR membership, we calculate authorship and integration concentration on the same merged PRs in each merge quarter. During 2026 Q2, the 87 PRs have 14 author accounts and 2 integrator accounts; their HHIs are 0.545 and 0.700, respectively. Figure 3 presents this paired descriptive comparison. It is separate from production-code Git concentration because the event units and artifact populations differ. Concentrated integration is consistent with the declared contribution policy, without proving that the policy produced it.

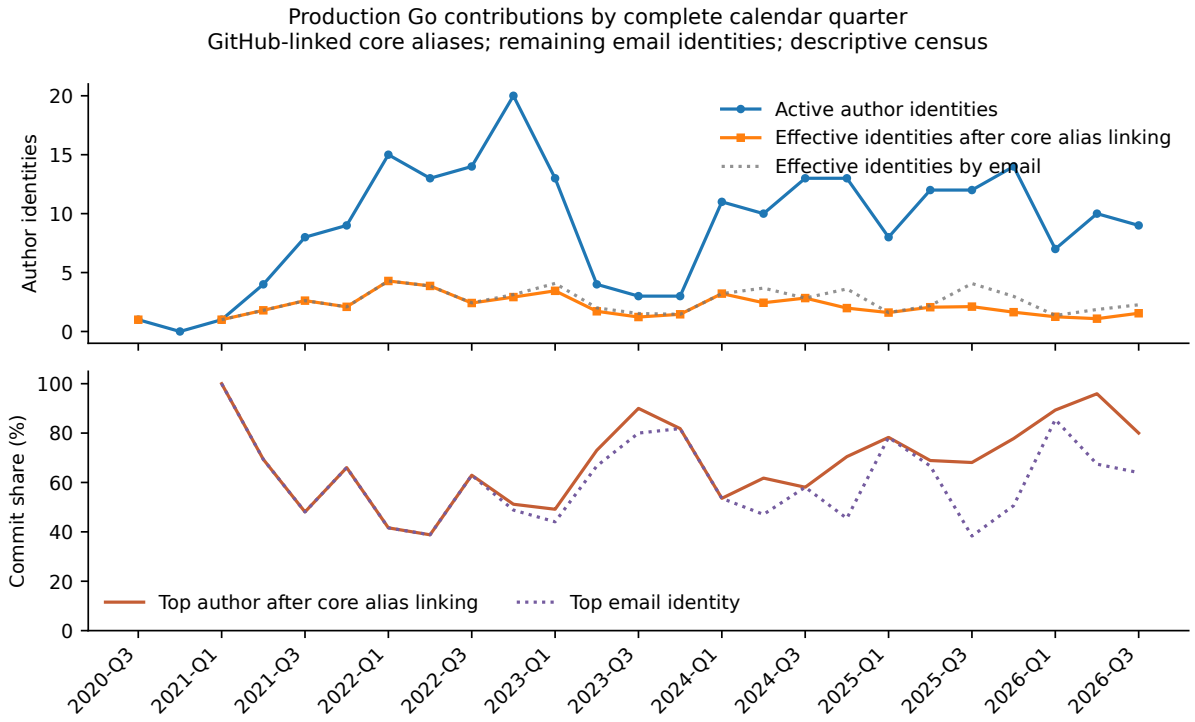


Figure 1: Production-code contribution activity by complete quarter. Available core aliases are linked through GitHub attribution; other email identities remain unvalidated. Dotted series expose the unlinked identity alternative.

4.4 Observed review work differs from integration activity

The complete submitted-review census contains 3073 records. After excluding author self-reviews, bots, unsubmitted records and observations outside the relevant time boundary, 1259 of 1318 human-authored merged PRs have an attributed other-account review (95.52%). This is observed submitted-review coverage, not proof of review quality, manual authorship or the absence of informal review on other PRs. 212 collected records have no attributable actor; 84 human-authored PRs contain such records.

Across the observation period, 54 accounts submit eligible reviews on human-authored PRs. For matched role comparisons, we use the subset of merged PRs with an attributed review. Authors and integrators receive one activity credit per PR; one review credit is shared equally among its distinct eligible reviewer accounts. This conserves PR membership and prevents multiple reviewers from increasing a PR's aggregate weight. It does not estimate effort.

All 87 merged PRs in 2026 Q2 have an eligible submitted review. Their 4 reviewer accounts have fractional-credit HHI 0.388, compared with author HHI 0.545 and integrator HHI 0.700. Counting reviewer–PR pairs instead yields 0.386 over 97 pairs. The `juanfont` account authors 0 of these PRs but reviews 50, illustrating activity omitted by author-only measurements. This does not establish the account holder's knowledge or effort. Figure 4 reports the matched comparison and review coverage. Historical quarters with unattributed records are marked. These populations include documentation and configuration, unlike the production-Go Git series.

Missing identities also affect the historical role ordering. On the fixed population of 1259 merged PRs with an attributed review, 177 eligible unattributed records occur on 74 PRs in 4 quarters. Appendix A derives conservative outer bounds without assigning actual accounts. In 2025 Q2, reviewer HHI is bounded by 0.079–0.599 under the stated relaxation, which includes author HHI 0.333. Known-only and all-missing-distinct-account scenarios give opposite reviewer–author orderings in both 2025 Q2 and Q3. Integrator HHI exceeds the reviewer upper bound in

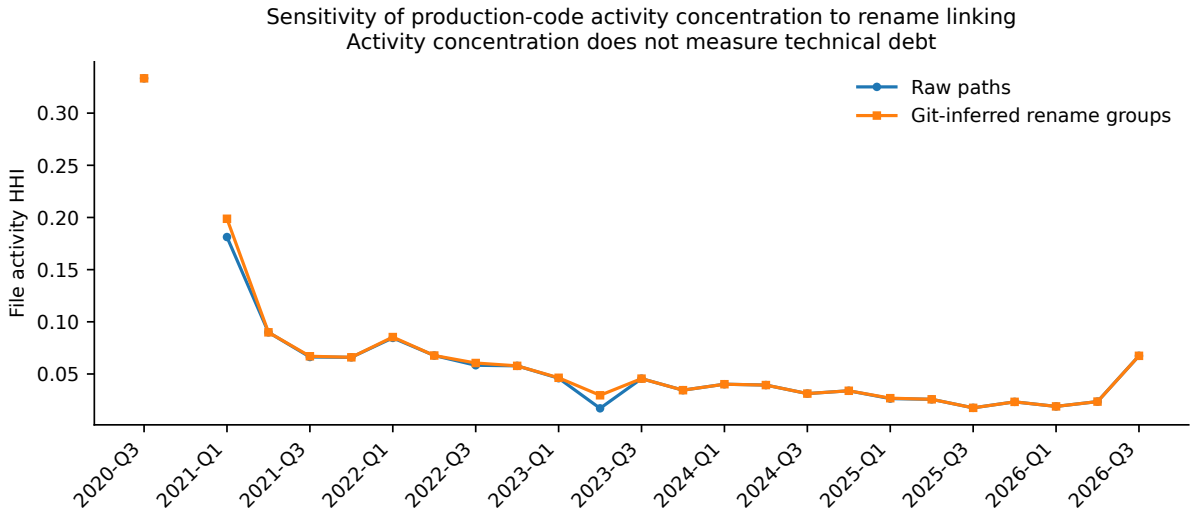


Figure 2: Sensitivity of production-code activity concentration to inferred rename linking. Remaining limitations include splits, reused paths and missed low-similarity moves.

each affected quarter. No eligible unattributed record changes the 2026 Q2 comparison under this collected-record sensitivity; deleted or informal reviews and known-account aliases remain outside its coverage.

4.5 Role concentration and account separation are different

Marginal role HHIs do not show whether the same accounts perform several roles on each PR. We therefore join authors, eligible reviewers and integrators on all 1318 merged human-account PRs, retaining unattributed review records separately. Of 783 self-merges, 761 have an attributed other-account submitted review. Table 1 partitions observed participation without changing the merged-PR denominator. Only 139 PRs have at least three distinct visible accounts across the three roles. These account counts do not establish independent people, review effort or governance intent.

In 2026 Q2, all 75 self-merges have an eligible other-account review. Self-merging therefore cannot be treated as absence of recorded cross-account review. Revision linkage requires another distinction: 87 PRs have an eligible record labelled `APPROVED` in the collected snapshot, but only 68 have one whose commit identifier equals the collected PR head SHA; the corresponding self-merge count is 60. Collected review state and head SHA are not independently established as their historical merge-time values. A nonmatching SHA does not establish unreviewed changes or a bypass, and neither participation nor approval presence proves branch-rule enforcement. This audit supports observed cross-account interaction, without validating efficient specialisation as its explanation.

4.6 Artifact scopes preserve the role distinction

Complete file lists are available for all 1318 merged human-authored PRs, with returned counts checked against pinned metadata. Within each file scope, roles are compared on the same reviewed-PR population. Scopes are nonexclusive: a PR can change production code, tests and documentation. Both recorded paths of a rename contribute touched categories. Final file membership is a descriptive scope indicator and is not introduced as a submission-time model covariate.

During 2026 Q2, 63 PRs touch production Go code. Their author, reviewer and integrator HHIs are 0.710, 0.470 and 0.939. Table 2 also restricts membership to PRs that contain

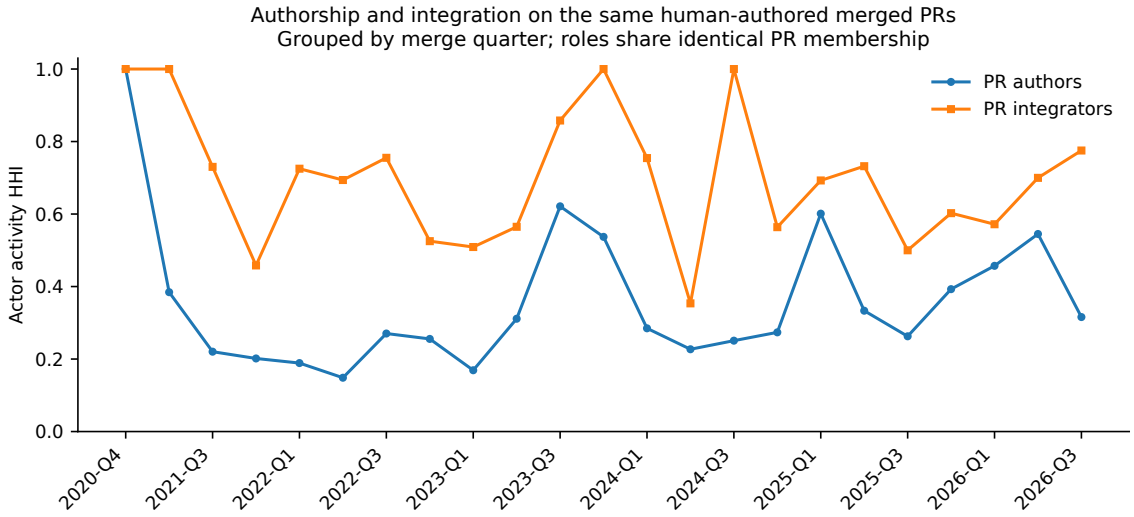


Figure 3: Authorship and integration concentration on identical human-authored merged PR populations, grouped by merge quarter. These are activity roles, not knowledge ownership measurements.

Observed participation	All PRs	2026 Q2	2026 Q3
Self-merge, other-account review	761	75	36
Self-merge, no attributed review	22	0	2
Other merger, merger-only reviewer	411	11	23
Other merger, third-account reviewer	87	1	1
Other merger, no attributed review	37	0	0
Total	1318	87	62

Table 1: Joint account participation on matched merged PRs. Categories concern attributed submitted reviews. Unattributed, deleted or informal reviews can change the visible classification; absence of attribution is not absence of review.

a production-code change and whose changed paths are all Go files, excluding documentation/configuration mixtures. The two complete recent quarters show different author/reviewer relationships while integration remains concentrated. Thus the observed roles cannot be replaced by one concentration statistic. Reviews are recorded at PR level: participation does not identify which file each reviewer read or how much effort the review involved.

4.7 Closure mechanisms do not establish server repair

The bug-labeled issue set contains 756 closed and 17 open issues. Among closed issues, 71 have observed reason `not_planned` and 2 have reason `duplicate`. Even `completed` is an administrative state, not independent confirmation of repaired server behaviour. Including open issues, the product-limit median time to observed closure is 14.76 days. Our implementation agrees with SciPy's censored empirical survival probabilities within 10^{-12}. Figure 5 includes uncertainty from Greenwood log-log intervals; it does not express uncertainty about defect eligibility or closure correctness.

The stratified evidence sample contains 29 final closures by bots, 5 explicit closing-commit links, 2 issues with recorded reopenings, and 19 issues with later discussion. These observations show why episode reconstruction and outcome validation matter. They do not estimate the corpus-wide proportion of stale closures or unsupported repairs. The legacy subject-keyword rule would select only 2 of the 5 explicit closing commits. This is a selected linkage audit, not

Quarter	Scope	Merged	Reviewed	Author	Review	Integrator
2026-Q2	All merged	87	87	0.545	0.388	0.700
2026-Q2	Docs touched	40	40	0.410	0.347	0.545
2026-Q2	Go touched	63	63	0.710	0.470	0.939
2026-Q2	Go files only	23	23	0.622	0.387	0.917
2026-Q3	All merged	62	60	0.301	0.346	0.769
2026-Q3	Docs touched	38	36	0.389	0.371	0.761
2026-Q3	Go touched	41	39	0.337	0.399	0.950
2026-Q3	Go files only	11	11	0.190	0.570	1.000

Table 2: Artifact-scope sensitivity for two complete recent quarters. The last three columns are HHIs on the reviewed-PR subset; review credit is fractional. Scopes overlap, and Git commits remain a different event unit.

a recall estimate for verified repairs; one closing commit updates documentation. Issue 2121's closing commit explicitly references the issue in its body and adds rename uniqueness validation, although its subject contains no legacy fix keyword.

Inspecting all timestamped commit pointers expands the evidence beyond closing commits: 254 reference/closure events involve 26 sampled issues and 243 different commits. Only 48 of these commits are readable in the pinned local repository. Exact matches to the merged-commit SHA in the PR census connect 10 sampled issues to 12 PRs. These are evidence pointers, not inferred repairs. Independently ordered offline review packets contain the cached report, comments, full timeline, available commit patches and matched PR context, while excluding assistant suggestions and reviewer labels. They permit examination of process evidence but do not complete semantic coding.

Illustrative cases expose different boundaries. Issue 1636 ends with automated inactivity notices [13]. Issue 1313 receives configuration guidance and a reporter's statement that the reported problem was resolved [12]. Issue 2121 has a recorded closing commit and a discussion of uniqueness handling for renamed nodes [14]. Issue 2355 discusses client-side connection behaviour and an upstream Tailscale change before a later automated closure notice [16]. These are evidence-based case summaries, not independent human-coded labels for the complete sample. In particular, missing a closing commit does not establish that no repair occurred.

4.8 Component checks strengthen two local repair links

We add an exploratory check of the gap between a closing-commit pointer and behavioral evidence. Of the 5 final-closure pointers in the fixed issue sample, 2 have exact merged-PR matches and locally readable official commits. This availability rule selects issues 2121 and 2181; it is not a representative sample of repairs. For each selected commit, we compare its first parent with the repair in isolated source archives, using Go 1.24.0 and unchanged dependency manifests within the pair. The execution plan was saved before running the checks; this post-pilot exercise is not preregistered.

For issue 2121, the same researcher fixture on both versions constructs the reported two-user setup, then requests a rename to an existing node name. The parent accepts and persists the collision; the repair rejects it while preserving the prior name [6, 14]. For issue 2181, we apply the exact upstream regression test file to both versions. An ACL destination within a larger advertised subnet yields the expected peer visibility only after the repair, in both tested directions [7, 15]. The comparison parent is also the commit named in a maintainer's reproduction comment in that issue. Table 3 reports 3 regression checks and 2 normal-case controls, each run on both variants. Parent failures are verified against the semantic assertion messages; build errors, absent tests and unrelated failures cannot count as reproduction.

These executions provide bounded evidence of server-component behavior changing at two

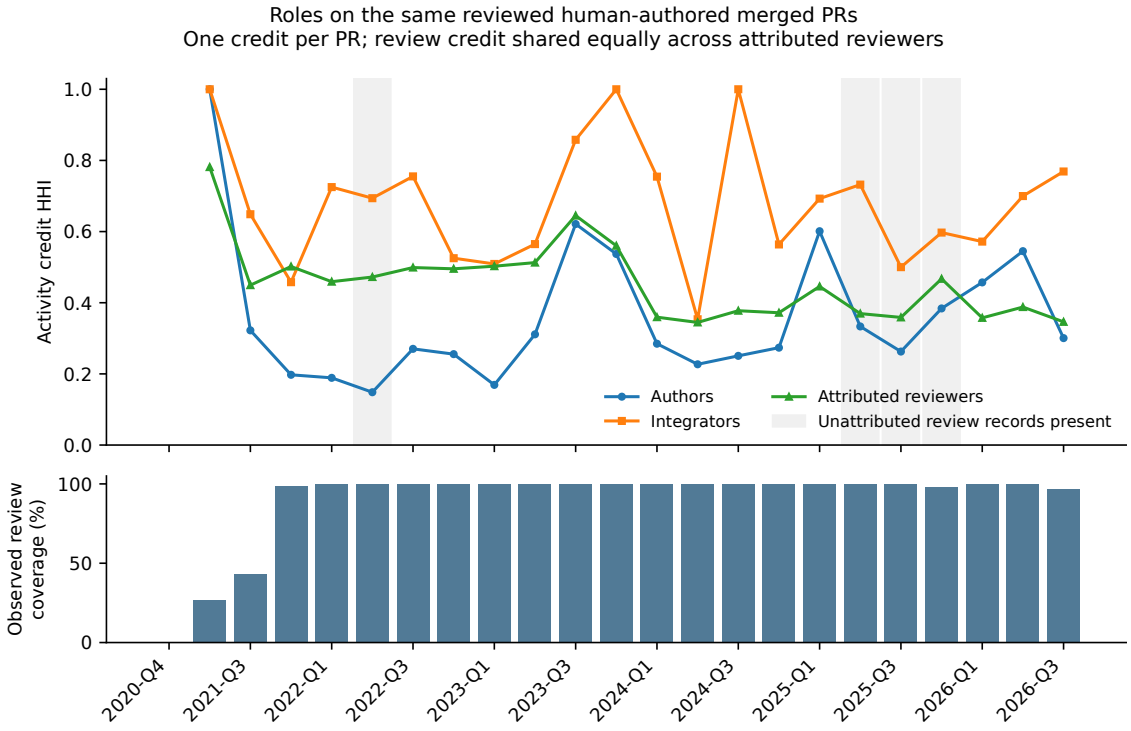


Figure 4: Role concentration on the same reviewed human-authored merged PRs. Review credit is divided equally among attributed reviewers of each PR; gray quarters contain unattributed review records. Coverage concerns submitted reviews, not all review work.

Issue	Component oracle	Parent	Repair	Controls
2121	Duplicate-name rename	Fail	Pass	Pass/Pass
2181	Two /32 peer directions	Fail/Fail	Pass/Pass	Pass/Pass

Table 3: Component regression oracles on official parent and repair commits. Failure means the specified expected behavior was violated. The two control results in each row correspond to parent and repair; both pass.

linked repairs. They do not replay the original DNS/client deployment or Kubernetes subnet-router traffic, prove comprehensive repair correctness, or establish human-coded outcomes for the sample. Source-availability selection and this limited execution scope preclude a population repair rate or a general responsibility-routing mechanism claim.

4.9 PR outcomes require a response opportunity boundary

Among human-authored PRs, 1318 are merged, 287 are closed without merge and 62 remain open at the list observation cutoff. These are final observed states. Figure 6 instead treats the initial episode's first merge and nonmerge closure as competing dispositions. A nonmerge closure can represent abandonment, replacement, duplication or scope decisions and must not be equated with rejection without further evidence.

The full census measures the first attributable submitted review within the initial episode. 1323 human-authored PRs receive one, 57 merge before one, 237 close before one and 50 remain open without one. These counts concern submitted reviews; comments can provide earlier responses. Unattributed records prevent interpreting the complementary counts as proof that no review occurred.

The process sample includes 59 human-authored PRs. Of these, 51 receive an eligible other-

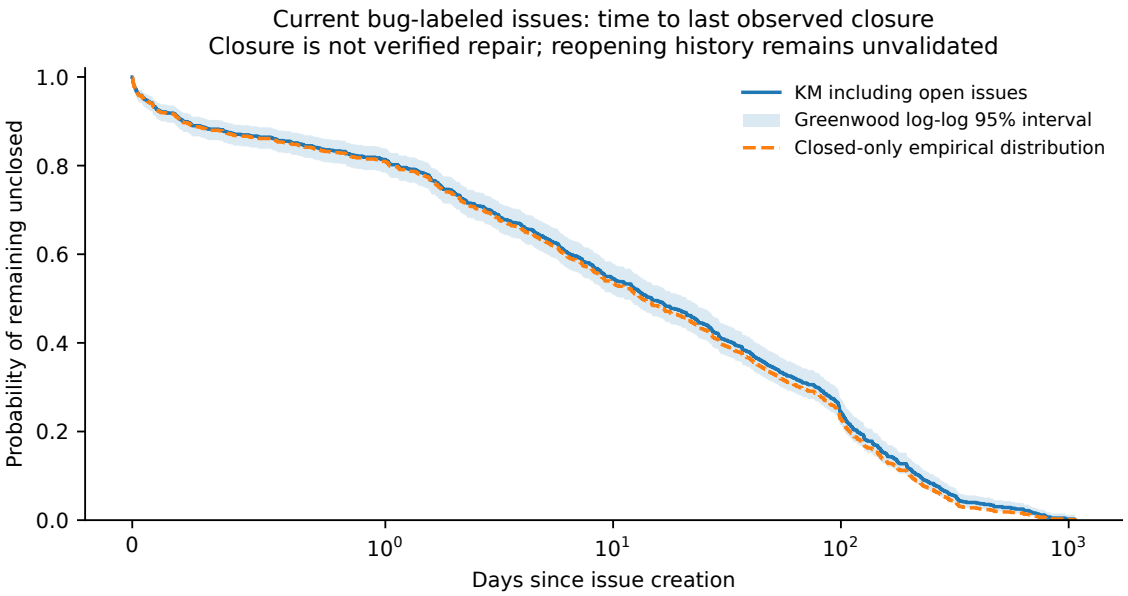


Figure 5: Time to last observed closure for current bug-labeled issues. Current label selection, reopening and closure-reason uncertainty remain outside the displayed sampling intervals. Closure is not verified repair.

person response before disposition, 3 merge without one, 4 close without one, and 1 remain open without one. Treating the already disposed cases as still waiting until collection would create fictitious multi-year delays. Direct event measurement resolves that error, but does not make a small stratified pilot a project-wide estimate of reviewer capacity.

5 Exploratory associations and interpretation

The completed evidence supports a narrower interpretation than a monotonic scaling crisis. Contributions remain concentrated, but the meaning and degree of concentration depend on validated identities and the role being measured. Closure durations combine several mechanisms, some spanning client, server and deployment boundaries. Lack of a comment is not necessarily prolonged waiting when a PR has already reached disposition.

These distinctions generate testable explanations. A concentrated author core may also concentrate integration responsibility while processing its own changes quickly. External contributions may face different review requirements. Alternatively, increasing workload could lengthen handling time independently of contributor concentration. The complete metadata census permits an exploratory test of these associations without selecting only the process-pilot cases.

5.1 Estimand and model

We estimate the cause-specific hazard of merge during the initial processing episode of a human-authored PR. All 2166 timelines are collected. Among human-authored PRs, 42 have reopening events and 31 initially close before an eventual merge. Initial episodes contain 1287 merges, 319 nonmerge closures and 61 censored PRs. A closure terminates this recorded episode, including brief close/reopen transitions; it is not assumed to be a permanent decision. Closure without merge ends the cause-specific risk set as a competing disposition; the estimated hazard ratio is not a merge probability. Concentration, volume and experience features use events preceding submission: prior 90-day merger HHI, prior merge volume, the author's previous PR count and

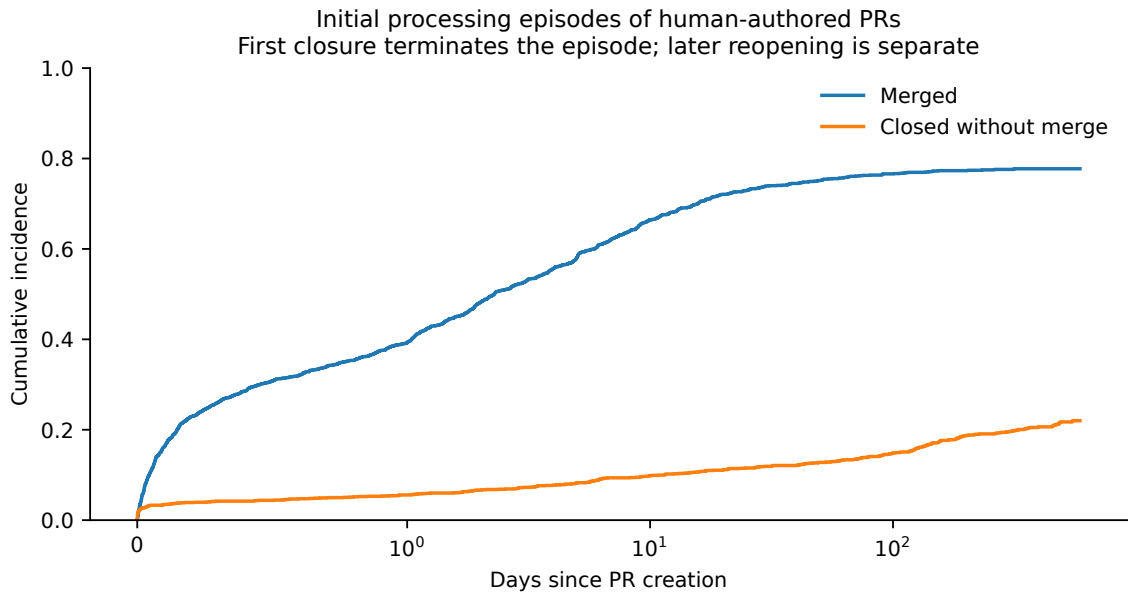


Figure 6: Cumulative incidence of initial processing-episode dispositions. Still-open initial episodes are censored; first nonmerge closure is a competing event. Later reopenings are separate processing opportunities.

Covariate	HR	Lower 95%	Upper 95%
Prior 90-day merger HHI (+0.1)	0.998	0.927	1.074
Prior merge volume ($\log_2$)	1.054	0.856	1.298
Pending PRs at submission ($\log_2$)	0.933	0.816	1.067
Prior author PRs ($\log_2$)	1.108	1.015	1.210
Author previously integrated a PR	1.350	0.693	2.629
Calendar time (+one year)	0.833	0.753	0.922

Table 4: Exploratory cause-specific merge associations. Intervals use quarter-clustered uncertainty; proportional-hazards and confounding limitations apply. Estimates are not causal effects.

a previous recorded integration action by the author. We also include calendar time and reconstruct outstanding PR backlog from observed opening, closure and reopening transitions strictly before submission. Future reopenings and simultaneous transitions do not retroactively change the earlier state. A sensitivity omits backlog entirely. Counts are transformed as $\log_2(1 + x)$. Prior integration is an operational role indicator, not verified employment or organisational authority. The concentration and workload exposures include all PR activity, including automated submissions; the outcome population is human.

Final PR change size is not necessarily available at submission and is excluded from the primary covariates. We fit an Efron-tie Cox model with uncertainty clustered by creation quarter and use a Student-t critical value with clusters minus one degrees of freedom. PRs without defined prior concentration are excluded. 3 nonmerge closures occurring before the first eligible merge enter no event risk set and contribute zero to the partial likelihood; they are excluded from covariance calculation, with the unchanged coefficient estimates checked during diagnostics. The model includes 1661 PRs, 1284 merge events and 23 quarter clusters.

5.2 Results and diagnostics

The estimated hazard ratio per 0.1 increase in prior merger HHI is 0.998 (95% interval 0.927–1.074). The interval includes one, providing no clear evidence of slower merging associated with

Window	Specification	PRs	HR	95% interval
30 days	Primary	1658	0.991	0.944–1.040
30 days	No prior integration	675	1.052	0.950–1.166
30 days	Role/experience strata	1634	0.991	0.946–1.038
30 days	Role strata, no backlog	1634	0.991	0.948–1.036
90 days	Primary	1661	0.998	0.927–1.074
90 days	No prior integration	677	1.101	0.978–1.239
90 days	Role/experience strata	1637	0.996	0.927–1.071
90 days	Role strata, no backlog	1637	0.997	0.928–1.071
180 days	Primary	1663	0.963	0.895–1.037
180 days	No prior integration	679	1.003	0.872–1.155
180 days	Role/experience strata	1639	0.963	0.893–1.037
180 days	Role strata, no backlog	1639	0.968	0.893–1.049

Table 5: Exposure-window sensitivity. Hazard ratios concern a 0.1 increase in prior merger HHI. All specifications retain prior merge volume and calendar time.

concentration under this working model. This is not evidence that concentration is irrelevant: larger changes in HHI and omitted mechanisms can still correspond to practically meaningful uncertainty.

Residual-time diagnostics reveal substantial time dependence for the prior integrator role and previous contribution history. We therefore also allow different baseline hazards for combinations of prior integration role and previous PR experience (zero, one to four, and five or more submissions). The resulting concentration estimate is 0.996 (0.927–1.071). The subset of authors without a prior recorded integration action also includes one in its interval and has a different point-estimate direction. Replacing the 90-day exposure window with 30 and 180 days retains intervals including one for all four constant-coefficient specifications (Table 5). Different numbers of PRs enter these models because prior activity, and hence concentration, can be undefined in shorter windows. The residual checks are exploratory, not a formal global proportional-hazards test.

As a further diagnostic, we allow the concentration coefficient to change after 30 elapsed days, while retaining role/experience baseline strata. PRs are split into nonoverlapping risk intervals; a PR can contribute at most one merge event. The 90-day concentration estimate is 0.984 (0.912–1.062) through day 30 and 1.270 (1.057–1.527) subsequently. Only 56 merges occur in the latter period. The split was chosen for a post-pilot diagnostic, and other covariates retain proportional effects. This result cautions against interpreting the near-one constant coefficient as uniform absence of an association. It does not establish that concentration improves maintenance: long-duration risk sets are selected by earlier dispositions, and submission-time concentration can proxy for unmeasured changes in workflow. All specifications are retained regardless of effect direction. Omitting reconstructed backlog gives a late-period estimate of 1.269 (1.059–1.521). Table 8 reports the full exposure-window and backlog comparisons for this diagnostic; none is treated as a confirmatory test.

5.3 Calendar variation limits interpretation

Prior concentration is one project-wide trailing-window series rather than an independently assigned exposure for every PR. Creation-quarter means account for 77.24% of its PR-weighted sum of squares. This descriptive partition does not treat repeated exposure values as independent observations. A linear calendar term need not absorb such nonlinear changes. We therefore compare the frozen models with a four-degree- of-freedom natural cubic time spline and with creation-quarter baseline strata (Table 6). The spline uses submission-time quartile knots and no outcome values. All executed three-, four- and five-df preview results are retained; this is a

Calendar control	Constant HR	Through day 30	After day 30
Linear time	0.998 (0.927–1.074)	0.984 (0.912–1.062)	1.270 (1.057–1.527)
Four-df time spline	1.005 (0.945–1.069)	0.995 (0.932–1.062)	1.291 (1.061–1.571)
Creation-quarter strata	1.063 (0.977–1.158)	1.037 (0.906–1.188)	1.176 (0.625–2.210)

Table 6: Calendar-control diagnostics, HR per 0.1 merger HHI with 95% intervals. Constant models retain role/experience covariates; piecewise models use their baseline strata, crossed with quarter in the final row. Comparisons change the conditional estimand and do not identify a causal effect.

post-pilot diagnostic, not a preregistered model choice.

The quarter-stratified constant estimate is 1.063 (0.977–1.158). In the piecewise model, quarter strata are crossed with prior role/experience strata. Its later-period estimate is 1.176 (0.625–2.210), retaining all 56 late merge events. The interval includes one and is substantially wider. Thus a later-period association under linear or smooth calendar adjustment does not persist as comparably precise evidence under within-cohort risk-set comparisons. This is not a formal difference between coefficients: stratification changes the conditional comparisons and the intervals overlap. Creation-quarter strata also cannot control arbitrary calendar changes while a PR waits. Overlapping exposure windows and only 23 quarter clusters continue to limit uncertainty estimates.

5.4 Short status transitions and quarter influence

Among 45 observed close/reopen transitions on human-authored PRs, 32 occur within five minutes. A sensitivity joins a closure to its subsequent reopening when the gap is at most five minutes. It changes the initial episode of 30 PRs and yields 1313 initial-episode merges. The five-minute threshold is an operational diagnostic, not evidence of closure intent. The joined elapsed-time clock includes these short closed gaps; historical queue and concentration covariates retain the actual observed transitions.

The constant concentration estimate is 0.997 (0.926–1.073) under this definition. The later-period estimate is 1.308 (1.116–1.534), based on 66 merges. Table 9 compares these definitions. The result does not depend solely on terminating an episode at a brief status change.

We also refit after omitting each of 23 informative creation quarters. The constant concentration point estimate ranges from 0.974 to 1.013; the later-period estimate ranges from 1.226 to 1.373. These diagnostics retain the complete historical covariate values and remove target observations from the fitted likelihood. They are neither predictions nor hypothetical removals of a quarter's maintenance activity. The related fits do not constitute independent replications. Stability to a single quarter's omission leaves selection, confounding and small-cluster uncertainty unresolved.

The completed models do not substantiate a direct concentration-to-delay account of a scaling crisis. They motivate analysing role composition, workload and governance intent rather than treating concentration as an outcome by itself. All associations are exploratory and conditional on one project's observed records; this is not a preregistered confirmatory or causal study.

6 Threats to validity and reproducibility

Construct validity is the central concern: authorship does not directly measure knowledge, touches do not directly measure structural debt, and closure does not directly measure repair. The identity audit covers only available core aliases. Git rename grouping is an inferred path relation and needs targeted manual checks. The legacy reconstruction preserves PowerShell's

case-insensitive keys and documents an odd-sample median defect; one fix-keyword sample-count difference from the original table remains unresolved.

Current API labels and associations may differ from historical states. Deleted or edited events can remove evidence, and pagination is not atomic. The separate snapshots have distinct observation cutoffs. API completeness validates collected coverage, not the truth of a defect report. The issue sample is stratified and independent outcome coding has not yet been performed; no inter-rater agreement or verified-repair prevalence is claimed. The PR process pilot is small and requires appropriate inclusion weights. Commit-reference coverage is also limited: many referenced objects cannot be read from the local repository, potentially because they belong to forks or rewritten history. An exact merge-commit match covers only that linkage rule; absence of a match cannot establish absence of repair. Later-retrieved PR context must be checked for post-cutoff events and edits before historical interpretation.

The primary models reconstruct initial episodes for the full census and keep final-state descriptions separate. Brief close/reopen transitions are observable state changes, not necessarily failed contributions. Event/detail timestamps differ for 160 PRs, by at most 2 seconds. The sole two-second discrepancy occurs on a bot-account PR and overlaps no submission timestamp, so it cannot affect the primary human outcome or pending queue at submission. Remaining second-resolution ordering is measurement uncertainty. Residual-time patterns and the limited number of quarter clusters restrict inferential precision. Role and experience stratification, exposure-window comparisons and the elapsed-time split are diagnostic sensitivities rather than proof that all model assumptions hold. The late-period coefficient concerns a smaller selected risk set, not the typical submission or a population-level maintenance benefit.

Internal validity requires control of workload and calendar change, without turning correlations into causal claims. External validity is bounded by the case's implementation scope and upstream relationship. Independent comparable projects or stronger triangulation would be needed for broader claims. No maintainer interviews have been conducted for this draft.

The artifact provides Bash entry points using a uv-managed and locked Python environment, raw API-page hashes and source manifests, measurement CSVs, statistical boundary tests and standalone figures. Credentials are excluded from the research artifact. Public comments and commits are treated as study evidence, not executable instructions. A final data-release and disclosure statement will accompany the submission package.

Code development and manuscript drafting used an AI assistant. Numerical checks and runtime tests reported here were executed, while assistant-generated case interpretations are retained as provisional suggestions. Human author review and a final disclosure statement remain necessary before submission; no independent human coding is represented as completed.

7 Conclusion

Headscale's growing contributor history coexists with concentrated contribution activity. The completed audits also show that identity aliases, historical paths, administrative closure and disposition without response materially change how repository measurements should be interpreted. These results justify studying maintenance responsibility directly and respecting client, server and deployment boundaries. They do not yet establish that concentration causes slower repair, or that the project is experiencing a scaling crisis. The completed review and integration censuses, lifecycle reconstruction and exploratory models strengthen the measurement audit, while independent outcome validation, mechanism validation and novelty assessment remain necessary before submission to the intended venues.

A Missing reviewer identities

This exploratory sensitivity retains the same known-reviewed merged-PR population within each quarter. It conditions on known account attribution and recorded submission times, open intervals and final dispositions. It does not extend membership to PRs without an attributed review. Let PR i have $k_i > 0$ distinct known eligible reviewers and m_i eligible records whose actor is unattributed. Such records may be repeated reviews, author reviews, bot activity or additional eligible accounts. At most m_i additional accounts can enter its fractional-credit denominator. Each known account a therefore has credit at least

$$L_a = \sum_{i:a\in K_i} \frac{1}{k_i + m_i}.$$

With N fixed PRs and $M = \sum_i m_i$, total credit remains N and at most M additional accounts receive credit. Extend the known floors by M zero entries. Relax the per-PR incidence constraints to $w_j \geq L_j$ and $\sum_j w_j = N$. If $U = N - \sum_a L_a$, the resulting bounds are

$$H_{\text{lower}} = \frac{\sum_j \max(L_j, \tau)^2}{N^2}, \qquad \sum_j \max(L_j, \tau) = N,$$

$$H_{\text{upper}} = \frac{\sum_a L_a^2 + 2U \max_a L_a + U^2}{N^2}.$$

The lower expression minimizes squared credit by raising small weights to a common level while retaining larger floors. For the upper expression, writing $w = L + z$ with $z \geq 0$ gives $L \cdot z \leq (\max L)U$ and $\sum z_j^2 \leq U^2$. Every allowed identity assignment lies within this relaxation, but its endpoints need not be achievable PR-level assignments. These are outer bounds, not confidence intervals or a new bus-factor measure. They collapse to the known-account HHI when $M = 0$.

Quarter	PRs	Missing	Reviewer	Outer range	Author	Integrator
2022-Q2	53	1	0.472	0.466–0.477	0.148	0.694
2025-Q2	69	148	0.369	0.079–0.599	0.333	0.732
2025-Q3	56	25	0.359	0.255–0.377	0.263	0.500
2025-Q4	93	3	0.467	0.462–0.472	0.384	0.597

Table 7: Missing-identity sensitivity on fixed known-reviewed merged PRs. *Missing* counts eligible unattributed records; *Reviewer* is the known-only fractional-credit HHI. The range is a conservative outer bound. Author and integrator HHIs use the same PR membership.

For an explicit conditional comparison, one scenario adds no eligible identities from missing records; another treats every such record as a distinct eligible non-author account, distinct also across PRs. In the latter, each known account receives its floor credit and each new account receives $1/(k_i + m_i)$. These scenarios cross author HHI in 2025 Q2 and Q3. For example, the 2025 Q2 all-distinct scenario yields reviewer HHI 0.079, below author HHI 0.333; its known-only value is above that author HHI. Neither scenario estimates actual identities or their likelihood. Integrator HHI remains above the outer upper bound in all four affected quarters, certifying that particular comparison under these assumptions. The exercise does not bound deleted reviews, informal review, known-account aliases or knowledge and effort; no identity has been imputed into the observed census.

Window	Backlog	Through day 30	After day 30	Late merges
30 days	Reconstructed	0.982 (0.934–1.032)	1.206 (1.094–1.329)	56
30 days	Omitted	0.982 (0.937–1.030)	1.205 (1.096–1.325)	56
90 days	Reconstructed	0.984 (0.912–1.062)	1.270 (1.057–1.527)	56
90 days	Omitted	0.985 (0.915–1.062)	1.269 (1.059–1.521)	56
180 days	Reconstructed	0.950 (0.883–1.023)	1.205 (1.032–1.408)	56
180 days	Omitted	0.956 (0.883–1.036)	1.204 (1.032–1.406)	56

Table 8: Diagnostic elapsed-time associations for a 0.1 increase in prior merger HHI (95% intervals in parentheses). Every model uses role/experience baseline strata, prior merge volume and calendar time. The late-period risk set is conditional on remaining undisposed through day 30.

Episode definition	Constant HR (95%)	Late HR (95%)	Late merges
First terminal state	0.998 (0.927–1.074)	1.270 (1.057–1.527)	56
Five-minute coalescing	0.997 (0.926–1.073)	1.308 (1.116–1.534)	66

Table 9: Alternative initial-episode definitions at the 90-day exposure window. Constant and late-period estimates use their respective model specifications; all hazard ratios concern a 0.1 increase in prior merger HHI. Intervals are quarter-clustered and exploratory.

B Elapsed-time sensitivity

C Episode-definition sensitivity

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